# When can AI discover adsorption kinetics?

Abdulhamid Batayhi

Department of Mechatronics Engineering, Faculty of Mechanical Engineering,
Yıldız Technical University, Istanbul, Türkiye

ORCID 0009-0007-0374-6755

A fixed-bed adsorption experiment reports concentrations and temperatures over time, most often at the column outlet. What an engineer needs from it is the uptake rate law, usually written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$. Equation-discovery methods offer to find the law itself, not only its coefficient: sparse regression over a library of candidate terms [1, 2] and its weak-form [3, 5], ensemble [6] and exact best-subset [9] variants, symbolic regression [10, 11], and neural networks embedded in the governing equations [12]. For adsorption, one study has done this from column data: Santana et al. trained a neural uptake term on noisy outlet breakthrough curves and read the LDF form back out, with the isotherm given exactly and at one rate and one noise level [12]. Repeated database searches found no other study that discovers an adsorption rate law from column data. Where such a result holds, and where it stops holding, is not known.

Part of the answer is classical. Near local equilibrium the mass-transfer coefficient is poorly determined by column data [13]: finite rates then act only through an apparent dispersion [15], the LDF coefficient can be extracted wrongly from outlet breakthrough [16], and most of the output variability belongs to the isotherm [17, 18]. Yet in the constant-pattern regime a lumped rate coefficient is recovered from breakthrough consistently with theory [19], so any limit must be conditioned on dispersion, isotherm shape, noise and isotherm error. Equation discovery has its own limits. On a single transient trajectory that settles onto a slow manifold, as a breakthrough run does, the candidate terms become nearly collinear and the data barely distinguish them [20–22]; noise induces phase transitions in what can be recovered [23, 56, 57]; and unique recovery needs conditions that non-chaotic dynamics generally lack [66]. These results say that discovery must fail somewhere for column data. None says where, for which methods, or whether it fails where the rate constant itself stops being identifiable.

The question matters most where the law is not the textbook one. In the water-harvesting framework MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, and the LDF approximation is least accurate there [58, 59]; a loading-dependent rate has been fitted by hand [44]. A method that could discover such a law from column data would be useful. Whether one can, at the noise and isotherm accuracy of a real experiment, is what this study tests.

We asked, in a pre-registered computational study, for which regimes the adsorption rate law can

be discovered from data a column experiment can produce, and whether discovery fails at the same boundary as the practical identifiability of the rate constant [47], or earlier. Ground truth came from a validated non-isothermal column model [50], for a Langmuir sorbent and a MOF-303-like step isotherm, over four decades of Damköhler number, three Péclet numbers, four noise levels and four levels of isotherm error, observed through interior probes of concentration, uptake and temperature, through probes without uptake, or at the outlet alone. Methods from each family above were run in their strongest available configuration, and identifiability was judged by profile likelihood on the same observations. The design, thresholds and verdict words were frozen before any method ran on the grid. **[PENDING: one or two sentences stating the corrected verdicts.]**

# Results

## Both isotherms pass the positive control

At the lowest Damköhler number, with no noise and no isotherm error, discovery and identifiability must both succeed in at least 90 % of the estimation replicates, or that isotherm is withheld from every verdict. Both isotherms passed: the best method recovered the LDF law in 135 of 135 replicates, and $k$ was identifiable in 135 of 135, for the Langmuir and for the step isotherm alike. (As first run, the step isotherm failed this check because of a search-range error in the identifiability code; the correction and both sets of verdicts are given in Supplementary Note 2.)

## Discovery fails long before identifiability

H2b compares, in each slice of the design (one isotherm, one Péclet number, one noise level and one isotherm error), the Damköhler number at which the best discovery method falls to 50 % success, $\mathrm{Da}^*_{\mathrm{disc}}$, with the number at which $k$ stops being identifiable in half the replicates, $\mathrm{Da}^*_{\mathrm{ident}}$. Of the 96 slices, 30 had both boundaries on the grid. In all 30 the verdict was "*discovery fails before identifiability*", so H2b's pre-registered overall word is the same (Fig. 2, Table 1). The gap was large everywhere: from 1.6 to 3.4 decades of Da, with a median of 2.3 decades for the Langmuir sorbent (20 slices) and 2.5 for the step isotherm (10 slices). Discovery failed between Da = 0.11 and 7.2, while $k$ stayed identifiable up to Da between 43 and 562. In every one of the 30 slices the best discovery method was the same, exact best-subset selection in the weak form.

Noise moved both boundaries, but not equally. With 0.5 % noise and an exact isotherm, discovery held to Da ≈ 5 to 7 and identifiability to between the two highest levels (Da = 316 and 1000). At 2 % noise the discovery boundary fell to Da ≈ 1, and at 5 % to Da ≈ 0.1 to 0.3, near the bottom of the grid, while the identifiability boundary stayed between Da ≈ 43 and 172. A small error in the isotherm hurt discovery more than noise did: at 0.5 % noise, an isotherm error of 0.5 % moved $\mathrm{Da}^*_{\mathrm{disc}}$ from about 6 to about 0.2, more than a decade, while $\mathrm{Da}^*_{\mathrm{ident}}$ moved from 562 to about 360.

The 66 slices without a verdict are not missing data. In each, one of the two probabilities did not cross 50 % anywhere on the grid, and the pre-registered words for that case are "*no boundary on the grid for one of the two*". **[PENDING from the figure data: how many of these never**

**reach 50 % discovery at any** Da **(expected at isotherm error** $\geq 2\,\%$**) and how many never fail (expected without noise).]** Many identifiability boundaries fell exactly between two adjacent levels of Da, where every replicate was identifiable at one level and none at the next; such a boundary is reported as lying between those levels (Methods). **[VERIFY from the figure data: all-to-none between adjacent levels for every zero-width interval.]**

# Methods

## Ground truth

Ground truth came from the finite-difference reference solver of the companion study [50]: a one-dimensional, axially dispersed, non-isothermal model of a packed column, verified there against closed-form solutions. Uptake followed one of two laws,

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{1}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{2}$$

where $s$ is the logistic function and $\theta$ is the fractional loading at the isotherm's step midpoint, so that under L2 the rate falls to 20 % of $k_0$ across the step. Two adsorbents were simulated with the same column geometry and flow (Supplementary Table 1): the companion study's favourable Langmuir sorbent, and water on a MOF-303-like framework, a cooperative (Type V) isotherm with its step at $P/P_0 = 0.15$, fed at 30 % relative humidity and 30 °C. The Damköhler number $\mathrm{Da} = k\,t_{\mathrm{stoich}}$ counts kinetic time constants per passage of the adsorption front, with $t_{\mathrm{stoich}}$ the stoichiometric time of the column (7200 s for Langmuir and 31 050 s for the step isotherm). Nine levels from $10^{-1}$ to $10^{3}$ at half-decade spacing therefore span $k$ from $1.39 \times 10^{-5}$ to $0.139\,\mathrm{s}^{-1}$ for Langmuir and from $3.22 \times 10^{-6}$ to $0.0322\,\mathrm{s}^{-1}$ for the step isotherm. Three axial-dispersion coefficients, $D_L \times 0.2$, $\times 1$ and $\times 5$ of the default, completed the grid. They give column Péclet numbers $\mathrm{Pe} = uL/D_L$ of about 470, 94 and 19, and the levels are named by these values below (the code and the pre-registration label them by the $D_L$ multiplier, so its "$\times 0.2$" level is the highest Pe). The grid has 81 solves at 400 cells and 400 stored times, each run until the outlet reached 95 % breakthrough, plus 9 isothermal solves for a control described below. The manifest that records every parameter is public.

## Observation models, noise and isotherm error

Every method and every identifiability calculation read the data through one function, so all saw the same corrupted observations. Three observation models were used: interior probes of $c$, $q$ and $T$ (O1), interior probes of $c$ and $T$ with $q$ hidden (O2), and the outlet $c(t)$ and $T(t)$ only (O3). O1 and O2 used 20 equally spaced probes; all models used 200 times spread evenly over the solved horizon. Noise was independent Gaussian with standard deviation $\sigma \in \{0, 0.5, 2, 5\}$ % of each channel's range. Methods were never given the true isotherm. They were given a measured one,

$q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$ with $\varepsilon \in \{0, 0.5, 2, 5\}$ %, where $g$ is a smooth random function of four Fourier modes on $[0,1]$, scaled to unit root-mean-square and drawn once per replicate. Every random draw was seeded by a hash of (cell, replicate, $\sigma$, $\varepsilon$, observation model), so each observation can be regenerated on its own. A replicate, one such observation of one cell, was the unit of analysis; each cell had 60.

## Discovery methods

Each method ran in the strongest configuration we could verify on known-answer systems before the pre-registration was frozen, and a method that could not be run that way was reported as such rather than replaced by a weaker one. The candidate library was $\{1, c, q, T, cq, q^2, c^2, q^*_{\mathrm{meas}}\}$; under L1 the true law is $k(q^*_{\mathrm{meas}} - q)$. The methods were sparse regression on a Savitzky–Golay estimate of $\mathrm{d}q/\mathrm{d}t$ (strong form) and on its weak, integral form with compactly supported test functions [3, 4], each with three solvers: sequentially thresholded least squares [1], a bagged ensemble with an inclusion threshold of 0.6 [6], and exact best-subset selection by enumeration with the Bayesian information criterion (M4, the objective of [9]); errors-in-variables best-subset regression given the true error levels (M5, an oracle); implicit sparse regression without any isotherm on an isothermal control (M6, SINDy-PI [8]); symbolic regression (M7, PySR [10]); a universal differential equation with a neural network in the uptake term, trained through the column model and read out by sparse and symbolic regression (M1, the pipeline of [12]); and regression of the state on $q$-free features, which returns the isotherm where uptake is at equilibrium (M9 [22]). On O2, uptake was first recovered from the gas-phase mass balance, and every O1 method was then run on it. M1 could be trained for at most 50 L-BFGS steps or 2 h per fit on the machine available, so it ran as a declared demonstration: Langmuir, Pe $\approx 94$, $\sigma = 2$ %, $\varepsilon = 0$, O3, the nine Da levels, one replicate. The best parameters seen during training were kept. As first run, a column solve that exhausted its step budget during training ended the fit with no readout (four of the nine fits); the corrected code keeps the best parameters in that case, and both runs are reported (Supplementary Note 2). A Kolmogorov–Arnold network [51] failed its known-answer test and its cost gate before the freeze and was not run on the grid. Method settings, solver thresholds and their known-answer tests are listed in Supplementary Note 1.

## Success

Discovery succeeded when the selected terms were exactly those of the true law, every sign was correct and, for L1, the ratio of the $q^*_{\mathrm{meas}}$ and $q$ coefficients was $-1$ within 10 %. The value of $k$ itself was not required. A method that raised an error counted as a failure; no replicate, cell or output was excluded.

## Identifiability

Practical identifiability of $k$ was judged by profile likelihood [47] with the true law known, on the same observations and with the same measured isotherm as the discovery methods: $k$ was practically identifiable when the 95 % profile interval (likelihood-ratio threshold 3.84) lay within $[k/2, 2k]$. On O1 the likelihood summed over all probes the fit of the known law, integrated exactly for piecewise-linear $q^*(t)$ along the observed $c$ and $T$. On O2 and O3 every point of the profile was a full column solve. The profile searched a coarse logarithmic grid of $k$ and refined it around the interval (41 points per pass on O1, 21 on O2 and O3, as pre-registered). Two corrections to the identifiability profiles were made during the study and are reported in full (Supplementary Note 2). First, on O2 and O3 a measured isotherm with $\varepsilon > 0$ could stall the stiff solver at large trial $k$ for days; each solve was therefore capped at 600 s and a capped solve was scored as a failed solve (amendment A5; two O3 rows had capped solves, none next to its interval). Second, the O1 grid first spanned $10^{-5}$ to $10\,\mathrm{s}^{-1}$, which excluded the step cells at Da = 0.1; it was widened to $10^{-7}$ to $10\,\mathrm{s}^{-1}$, a peak on the grid edge was made an error, and all O1 rows were recomputed (amendment A6). Before that recomputation, the O1 refinement was found to use 21 points per pass instead of the pre-registered 41 and was set to 41, so every corrected O1 row uses the declared settings.

## Boundaries and verdicts

For each slice the boundary $\mathrm{Da}^*$ was the 50 % crossing of an unpenalised logistic regression of replicate outcomes on $\log_{10}\mathrm{Da}$, with a percentile bootstrap interval (2000 resamples within each Da level). No boundary was reported, and no verdict issued, when the outcomes did not cross 50 % inside the tested range; the range was reported instead. Before the freeze the estimator's coverage on planted boundaries was 0.94 for a nominal 0.95, with a bias of +0.012 decade. The best discovery method of each slice was chosen on replicates 0–14 and its boundary estimated on replicates 15–59, which removes the inflation that choosing and estimating on the same replicates causes (0.07 decade in simulation); the identifiability boundary was estimated on the same 45 replicates. M5, which knows the true error levels, took no part in this choice.

Three hypotheses were confirmatory, each with verdict words fixed in advance. H2a: success collapses onto one curve in $R = \delta/\varepsilon_{\mathrm{eff}}$, the driving force relative to the combined error, with the spread of $R^*$ across conditions under half a decade. H2b: on O1, for each slice ($\sigma$, $\varepsilon$, isotherm, Pe; 96 in all), the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ was called "*discovery fails before identifiability*" at $-0.25$ decade or less, "*discovery outlives identifiability*" at +0.25 or more, and "*discovery fails with identifiability*" in between; the hypothesis-level verdict was the word held by at least two thirds of the slices with a verdict. H2c: errors-in-variables regression moves the discovery boundary upwards by at least 0.25 decade in at least two thirds of the slices at a given $\varepsilon > 0$. With 45 estimation replicates the H2b rule returned a wrong directional verdict with probability 0.001 when there was no gap, and the correct one with probability 0.965 at a gap of 0.40 decade, the smallest effect of interest. A positive control guarded every verdict: in each isotherm's easiest cells (lowest Da, $\sigma = \varepsilon = 0$, O1) the best method had to succeed and $k$ had to be identifiable in at least 90 % of

replicates, or that isotherm's verdicts were withheld. O2 and O3 were reported descriptively, and H2d, whether the loading-dependent rate of L2 can be discovered, was exploratory.

### Pre-registration and amendments

The design was frozen on 11 October 2026 (commit `7dcc2fe`) before any discovery method ran on any grid cell, and every confirmatory driver refused to run unless the pre-registration matched its latest freeze commit. Six dated amendments are appended to it. A1 to A4 were made before the grid ran: 60 replicates split into selection and estimation sets, the smallest effect of interest, the oracle kept out of the pooled verdicts, Pe as a slicing axis, errors counted as failures, the decision to run the grid before peer review, and the M1 demonstration. A5 (the solve cap) was made while the grid ran, and A6 (the O1 grid range) after the verdicts were first computed. Both as-run and corrected verdicts are public, and the paper reports both (Supplementary Note 2).

### Exploratory analyses

Three analyses were declared in writing before they ran and are reported as exploratory: M1 trained at the scale a universal differential equation needs (E1), the Kolmogorov–Arnold network on its declared subgrid once the machine was free (E2), and how often an interval that meets $[k/2, 2k]$ misses the true $k$ (E3).

### Software and computation

The grid ran on a single computer without a graphics processor, one job at a time, from 12 October to 1 November 2026. Package versions are pinned in the repository.

### Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

## Data availability

The ground-truth manifest, every results file, the run log and the as-run and corrected verdicts are public in the study repository and archived on Zenodo (doi:10.5281/zenodo.23325994). The ground-truth fields are regenerated from the manifest by `p2_generate.py`.

## Code availability

All code is public at https://github.com/abdulhamidbatayhi123/ai-kinetics-discoverability under the MIT licence and archived with the data. The solver is imported unchanged from the companion study's archive (doi:10.5281/zenodo.23323685).

## Author contributions

A.B. is the sole author and conceived and designed the study. The use of AI tools is described in Methods.

## Competing interests

The author declares no competing interests.

## References


[1] Steven L. Brunton, Joshua L. Proctor, J. Nathan Kutz, Discovering governing equations from data by sparse identification of nonlinear dynamical systems, Proceedings of the National Academy of Sciences, vol. 113, 3932-3937 (2016). doi:10.1073/pnas.1517384113; arXiv:1509.03580.

[2] Samuel H. Rudy, Steven L. Brunton, Joshua L. Proctor, J. Nathan Kutz, Data-driven discovery of partial differential equations, Science Advances, vol. 3, e1602614 (2017). doi:10.1126/sciadv.1602614; arXiv:1609.06401.

[3] Daniel A. Messenger, David M. Bortz, Weak SINDy: Galerkin-Based Data-Driven Model Selection, Multiscale Modeling & Simulation, vol. 19, 1474-1497 (2021). doi:10.1137/20M1343166; arXiv:2005.04339.

[4] Daniel A. Messenger, David M. Bortz, Weak SINDy for partial differential equations, Journal of Computational Physics, vol. 443, 110525 (2021). doi:10.1016/j.jcp.2021.110525; arXiv:2007.02848.

[5] Patrick A. K. Reinbold, Daniel R. Gurevich, Roman O. Grigoriev, Using noisy or incomplete data to discover models of spatiotemporal dynamics, Physical Review E, vol. 101, 010203 (2020). doi:10.1103/PhysRevE.101.010203; arXiv:1911.03365.

[6] U. Fasel, J. N. Kutz, B. W. Brunton, S. L. Brunton, Ensemble-SINDy: Robust sparse model discovery in the low-data, high-noise limit, with active learning and control, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 478, 20210904 (2022). doi:10.1098/rspa.2021.0904; arXiv:2111.10992.

[7] Seth M. Hirsh, David A. Barajas-Solano, J. Nathan Kutz, Sparsifying priors for Bayesian uncertainty quantification in model discovery, Royal Society Open Science, vol. 9, 211823 (2022). doi:10.1098/rsos.211823; arXiv:2107.02107.

[8] Kadierdan Kaheman, J. Nathan Kutz, Steven L. Brunton, SINDy-PI: a robust algorithm for parallel implicit sparse identification of nonlinear dynamics, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 476, 20200279 (2020). doi:10.1098/rspa.2020.0279; arXiv:2004.02322.

[9] Dimitris Bertsimas, Wes Gurnee, Learning sparse nonlinear dynamics via mixed-integer optimization, Nonlinear Dynamics, vol. 111, 6585-6604 (2023). doi:10.1007/s11071-022-08178-9; arXiv:2206.00176.

[10] Miles Cranmer, Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl, arXiv preprint arXiv:2305.01582 (2023). arXiv:2305.01582.

[11] Silviu-Marian Udrescu, Max Tegmark, AI Feynman: A physics-inspired method for symbolic regression, Science Advances, vol. 6, eaay2631 (2020). doi:10.1126/sciadv.aay2631; arXiv:1905.11481.

[12] Vinicius V. Santana, Erbet Costa, Carine M. Rebello, Ana Mafalda Ribeiro, Christopher Rackauckas, Idelfonso B.R. Nogueira, Efficient hybrid modeling and sorption model discovery for non-linear advection-diffusion-sorption systems: A systematic scientific machine learning approach, Chemical Engineering Science, vol. 282, 119223 (2023). doi:10.1016/j.ces.2023.119223; arXiv:2303.13555.

[13] Albert J. Valocchi, Validity of the Local Equilibrium Assumption for Modeling Sorbing Solute Transport Through Homogeneous Soils, Water Resources Research, vol. 21, 808-820 (1985). doi:10.1029/wr021i006p00808.

[14] Scott K. Hansen, Velimir V. Vesselinov, Local equilibrium and retardation revisited, Groundwater, vol. 56, 109-117 (2018). doi:10.1111/gwat.12566; arXiv:1703.03087.

[15] Konstantinos Katsoulas, Monica Tirapelle, Eva Sorensen, Luca Mazzei, On the apparent dispersion coefficient of the equilibrium dispersion model: An asymptotic analysis, Journal of Chromatography A, vol. 1708, 464345 (2023). doi:10.1016/j.chroma.2023.464345.

[16] James C. Knox, Armin D. Ebner, M. Douglas LeVan, Robert F. Coker, James A. Ritter, Limitations of Breakthrough Curve Analysis in Fixed-Bed Adsorption, Industrial & Engineering Chemistry Research, vol. 55, 4734-4748 (2016). doi:10.1021/acs.iecr.6b00516.

[17] Adam Ward, Ronny Pini, Integrated uncertainty quantification and sensitivity analysis of single-component dynamic column breakthrough experiments, Adsorption, vol. 28, 161-183 (2022). doi:10.1007/s10450-022-00361-z.

[18] W. Taylor, J.D. Herman, V.L. Morales, Global sensitivity of laboratory-scale groundwater reactive transport simulations for sorption and degradation: Breakthrough dynamics across Péclet–Damköhler regimes, Journal of Contaminant Hydrology, vol. 283, 105043 (2026). doi:10.1016/j.jconhyd.2026.105043.

[19] Kanji Miyabe, Georges Guiochon, Determination of the lumped mass transfer rate coefficient by frontal analysis, Journal of Chromatography A, vol. 890, 211-223 (2000). doi:10.1016/s0021-9673(00)00537-9.

[20] S. Stanhope, J. E. Rubin, D. Swigon, Identifiability of Linear and Linear-in-Parameters Dynamical Systems from a Single Trajectory, SIAM Journal on Applied Dynamical Systems, vol. 13, 1792-1815 (2014). doi:10.1137/130937913.

[21] Matteo Gallo, Fabio Anselmi, Paolo Lazzari, The dynamics of discoverability: how trajectories and priors shape equation recovery, arXiv preprint arXiv:2607.18490v2 (2026). arXiv:2607.18490.

[22] Diemen Delgado-Cano, Erick Kracht, Urban Fasel, Benjamin Herrmann, SINDy on slow manifolds, arXiv preprint arXiv:2507.00747 (2025). arXiv:2507.00747.

[23] Andrei A. Klishin, Joseph Bakarji, J. Nathan Kutz, Krithika Manohar, Statistical mechanics of dynamical system identification, Physical Review Research, vol. 7, 033181 (2025). doi:10.1103/4d98-tdlp; arXiv:2403.01723.

[24] Ismaila Muhammed, Dimitris M. Manias, Dimitris A. Goussis, Haralampos Hatzikirou, Data-driven identification of biological systems using multi-scale analysis, PLOS Computational Biology, vol. 21, e1013193 (2025). doi:10.1371/journal.pcbi.1013193.

[25] Siddharth Prabhu, Nick Kosir, Mayuresh V. Kothare, Srinivas Rangarajan, Derivative-Free Domain-Informed Data-Driven Discovery of Sparse Kinetic Models, Industrial & Engineering Chemistry Research, vol. 64, 2601-2615 (2025). doi:10.1021/acs.iecr.4c02981.

[26] Juntao Huang, Yizhou Zhou, Wen-An Yong, Data-driven discovery of multiscale chemical reactions governed by the law of mass action, Journal of Computational Physics, vol. 448, 110743 (2022). doi:10.1016/j.jcp.2021.110743; arXiv:2101.06589.

[27] Charles B. Delahunt, J. Nathan Kutz, A toolkit for data-driven discovery of governing equations in high-noise regimes, IEEE Access, vol. 10, 31210-31234 (2022). doi:10.1109/access.2022.3159335; arXiv:2111.04870.

[28] Alan A. Kaptanoglu, Lanyue Zhang, Zachary G. Nicolaou, Urban Fasel, Steven L. Brunton, Benchmarking sparse system identification with low-dimensional chaos, Nonlinear Dynamics, vol. 111, 13143-13164 (2023). doi:10.1007/s11071-023-08525-4; arXiv:2302.10787.

[29] Idelfonso B. R. Nogueira, Vinicius V. Santana, Ana M. Ribeiro, Alírio E. Rodrigues, Using scientific machine learning to develop universal differential equation for multicomponent adsorption separation systems, The Canadian Journal of Chemical Engineering, vol. 100, 2279-2290 (2022). doi:10.1002/cjce.24495.

[30] Jesper Frandsen, Vinicius V. Santana, Peter Jul-Rasmussen, Idelfonso B.R. Nogueira, Jakob K. Huusom, Krist V. Gernaey, Jens Abildskov, A systematic screening of neural network-based hybrid models of adsorption in chromatography processes, AIChE Journal, vol. 71, e70045 (2025). doi:10.1002/aic.70045.

[31] Arthur Jessop, Mohammed Alsubeihi, Ben Moseley, Ashwin Kumar Rajagopalan, Differentiable Hybrid Modelling for Learning and Optimising Chemical Transport Processes from Experimental Data, arXiv preprint arXiv:2609.04011 (2026). arXiv:2609.04011.

[32] Xiao Xu, Marta D'Elia, Christian Glusa, John T. Foster, Machine-learning of nonlocal kernels for anomalous subsurface transport from breakthrough curves, arXiv preprint arXiv:2201.11146 (2022). arXiv:2201.11146.

[33] Jay Shukla, Xiaohui Qu, Zubin Darbari, Marija Iloska, J. Anibal Boscoboinik, Qin Wu, Discovering CO adsorption and desorption pathways from chemical reaction neural network modeling of transient kinetics spectroscopy, The Journal of Physical Chemistry Letters, vol. 16, 3562-3570 (2025). doi:10.1021/acs.jpclett.5c00665.

[34] Timothy Praditia, Matthias Karlbauer, Sebastian Otte, Sergey Oladyshkin, Martin V. Butz, Wolfgang Nowak, Learning Groundwater Contaminant Diffusion-Sorption Processes With a Finite Volume Neural Network, Water Resources Research, vol. 58, e2022WR033149 (2022). doi:10.1029/2022WR033149.

[35] Sai Gokul Subraveti, Zukui Li, Vinay Prasad, Arvind Rajendran, Can a computer "learn" nonlinear chromatography?: Physics-based deep neural networks for simulation and optimization of chromatographic processes, Journal of Chromatography A, vol. 1672, 463037 (2022). doi:10.1016/j.chroma.2022.463037.

[36] Stefania Scheurer, Riccardo Frenner, Tim Brünnette, Sergey Oladyshkin, Wolfgang Nowak, Efficient confidence interval computation for physics-aware machine learning of diffusion-sorption models, Frontiers in Water, vol. 8, 1813791 (2026). doi:10.3389/frwa.2026.1813791.

[37] Reza Haghpanah, Danny Shade, Fitting Adsorption Isotherms with Symbolic Regression, Industrial & Engineering Chemistry Research, vol. 62, 22141-22148 (2023). doi:10.1021/acs.iecr.3c02900.

[38] Samiha Sharlin, Tyler R. Josephson, In Context Learning and Reasoning for Symbolic Regression with Large Language Models, arXiv preprint arXiv:2410.17448 (2024). arXiv:2410.17448.

[39] Charles Fox, Neil Tran, Nikki Nacion, Samiha Sharlin, Tyler R. Josephson, Incorporating background knowledge in symbolic regression using a computer algebra system, Machine Learning: Science and Technology, vol. 5, 025057 (2024). doi:10.1088/2632-2153/ad4a1e; arXiv:2301.11919.

[40] Miguel Ángel de Carvalho Servia, Ilya Orson Sandoval, King Kuok (Mimi) Hii, Klaus Hellgardt, Dongda Zhang, Ehecatl Antonio del Rio Chanona, The automated discovery of kinetic rate models – methodological frameworks, Digital Discovery, vol. 3, 954-968 (2024). doi:10.1039/d3dd00212h; arXiv:2301.11356.

[41] Miguel Ángel de Carvalho Servia, Ilya Orson Sandoval, King Kuok, Hii, Klaus Hellgardt, Dongda Zhang, Ehecatl Antonio del Rio Chanona, Constraint-Guided Symbolic Regression for Data-Efficient Kinetic Model Discovery, arXiv preprint arXiv:2507.02730 (2025). arXiv:2507.02730.

[42] Torkel E Loman, Ruth E Baker, Functional and parametric identifiability for universal differential equations applied to chemical reaction networks, arXiv preprint arXiv:2510.14140 (2025). arXiv:2510.14140.

[43] Konstantinos Katsoulas, Federico Galvanin, Luca Mazzei, Eva Sorensen, A Diagnostic Procedure for Identifying Isotherm Models in Liquid Chromatography, Industrial & Engineering Chemistry Research, vol. 65, 1277-1292 (2026). doi:10.1021/acs.iecr.5c03704.

[44] Ahmed E. Abu EL-Maaty, Mahmoud A. Abdalla, Mohamed Essalhi, Mahmoud M. Abdelnaby, Morsi M. Mahmoud, Mohamed A. Habib, Mohamed Antar, Rached Ben-Mansour, Kinetics of Water Adsorption in Metal-Organic Framework(MOF-303) for Adsorption Cooling Application, Energy Conversion and Management: X, vol. 24, 100694 (2024). doi:10.1016/j.ecmx.2024.100694.

[45] Yuta Sakanaka, Shotaro Hiraide, Iori Sugawara, Hajime Uematsu, Shogo Kawaguchi, Minoru T. Miyahara, Satoshi Watanabe, Generalised analytical method unravels framework-dependent kinetics of adsorption-induced structural transition in flexible metal–organic frameworks, Nature Communications, vol. 14, 6862 (2023). doi:10.1038/s41467-023-42448-3.

[46] Reid A. Coyle, Shyam Chand Pal, Peter Walther, Saeun Park, Bin Feng, Zhiling Zheng, Sustainable metal–organic framework water harvesters in the artificial intelligence era, ACS Materials Letters, vol. 8, 2391-2400 (2026). doi:10.1021/acsmaterialslett.6c00580; arXiv:2605.29179.

[47] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, J. Timmer, Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood, Bioinformatics, vol. 25, 1923-1929 (2009). doi:10.1093/bioinformatics/btp358.

[48] Lloyd Fung, Overcoming error-in-variable or stiffness problem in SINDy-like data-driven model discovery, ACM Transactions on AI for Science, 3831701 (2026). doi:10.1145/3831701.

[49] David M. Bortz, Daniel A. Messenger, Vanja Dukic, Direct estimation of parameters in ODE models using WENDy: weak-form estimation of nonlinear dynamics, Bulletin of Mathematical Biology, vol. 85, 110 (2023). doi:10.1007/s11538-023-01208-6; arXiv:2302.13271.

[50] A. Batayhi, The coefficient map, not the basis: what limits neural-operator and reduced-order surrogates on unseen adsorbents, companion manuscript (2026); code, results and pre-registrations archived at Zenodo, version v1.0.4, doi:10.5281/zenodo.23312204.

[51] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, Max Tegmark, KAN: Kolmogorov-Arnold Networks, ICLR 2025 (arXiv 2024). arXiv:2404.19756.

[52] Kevin Slote, Jeremie Fish, Erik Bollt, KANDy: Kolmogorov-Arnold Networks and Dynamical System Discovery, arXiv preprint arXiv:2602.20413 (2026). arXiv:2602.20413.

[53] Enzo Nicolas Spotorno, Josafat Leal Filho, Antonio Augusto Medeiros Frohlich, Empirical Stability Analysis of Kolmogorov-Arnold Networks in Hard-Constrained Recurrent Physics-Informed Discovery, arXiv preprint arXiv:2602.09988 (2026). arXiv:2602.09988.

[54] Gene H. Golub, Charles F. Van Loan, An analysis of the total least squares problem, SIAM Journal on Numerical Analysis, vol. 17(6), 883-893 (1980). doi:10.1137/0717073.

[55] G. H. Golub, Alan Hoffman, G. W. Stewart, A generalization of the Eckart-Young-Mirsky matrix approximation theorem, Linear Algebra and its Applications, vol. 88-89, 317-327 (1987). doi:10.1016/0024-3795(87)90114-5.

[56] Oscar Fajardo-Fontiveros, Ignasi Reichardt, Harry R. De Los Ríos, Jordi Duch, Marta Sales-Pardo, Roger Guimerà, Fundamental limits to learning closed-form mathematical models from data, Nature Communications, vol. 14, 1043 (2023). doi:10.1038/s41467-023-36657-z.

[57] Daniel A Messenger, David M Bortz, Asymptotic consistency of the WSINDy algorithm in the limit of continuum data, IMA Journal of Numerical Analysis, vol. 45, 3264–3312 (2025). doi:10.1093/imanum/drae086.

[58] Jon Hastings, Thomas Lassitter, Zhiling Zheng, Saumil Chheda, J. Ilja Siepmann, Laura Gagliardi, Omar M. Yaghi, T. Grant Glover, High-Temperature Water Adsorption Isotherms and Ambient Temperature Water Diffusion Rates on Water Harvesting Metal–Organic Frameworks, The Journal of Physical Chemistry C, vol. 128, 11328–11339 (2024). doi:10.1021/acs.jpcc.4c01733.

[59] Thomas Lassitter, Nikita Hanikel, Dennis J. Coyle, Mohammad I. Hossain, Bryce Lipinski, Michael O'Brien, David B. Hall, Jon Hastings, Juan Borja, Travis O'Neil, S. Ephraim

Neumann, David R. Moore, Omar M. Yaghi, T. Grant Glover, Mass transfer in atmospheric water harvesting systems, Chemical Engineering Science, vol. 285, 119430 (2024). doi:10.1016/j.ces.2023.119430.

[60] Chunye Ma, Jiawang Li, Guang Wang, Zhigang Li, Wei Su, Yanguang Zhou, Microscopic Kinetics of Water Adsorption in Metal–Organic Frameworks, Nano Letters, vol. 25, 5778–5783 (2025). doi:10.1021/acs.nanolett.5c00343.

[61] Max D. Champneys, Timothy J. Rogers, BINDy: Bayesian identification of nonlinear dynamics with reversible-jump Markov-chain Monte Carlo, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 481, 20240620 (2025). doi:10.1098/rspa.2024.0620.

[62] Kai Ruan, Yilong Xu, Ze-Feng Gao, Yang Liu, Yike Guo, Ji-Rong Wen, Hao Sun, Discovering physical laws with parallel symbolic enumeration, Nature Computational Science, vol. 6, 53–66 (2025). doi:10.1038/s43588-025-00904-8.

[63] David Büchner, Ronny Pini, Unary and Binary Dynamic Column Breakthrough Experiments with Carbon Dioxide and Nitrogen Imaged by X-ray Computed Tomography, Langmuir, vol. 42, 14716–14732 (2026). doi:10.1021/acs.langmuir.6c00250.

[64] Jialu Wang, Alexander W. Dowling, Pyomo.DOE: An open-source package for model-based design of experiments in Python, AIChE Journal, vol. 68, e17813 (2022). doi:10.1002/aic.17813.

[65] Cristina Cornelio, Sanjeeb Dash, Vernon Austel, Tyler R. Josephson, Joao Goncalves, Kenneth L. Clarkson, Nimrod Megiddo, Bachir El Khadir, Lior Horesh, Combining data and theory for derivable scientific discovery with AI-Descartes, Nature Communications, vol. 14, 1777 (2023). doi:10.1038/s41467-023-37236-y.

[66] Zakhar Shumaylov, Peter Zaika, Philipp Scholl, Gitta Kutyniok, Lior Horesh, Carola-Bibiane Schönlieb, When is a System Discoverable from Data? Discovery Requires Chaos, arXiv preprint arXiv:2511.08860 (2025). arXiv:2511.08860.

[67] Nic Rummel, Daniel A. Messenger, Stephen Becker, Vanja Dukic, David M. Bortz, WENDy for Nonlinear-in-Parameters ODEs, arXiv preprint arXiv:2502.08881 (2025). arXiv:2502.08881.